Multitemporal and Multispectral Data Fusion for Super-Resolution of Sentinel-2 Images

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***Abstract*— Multispectral Sentinel-2 (S-2) images are a valuable source of Earth observation data; however, spatial resolution of their spectral bands limited to 10-, 20-, and 60-m ground sampling distance (GSD) remains insufficient in many cases. This problem can be addressed with super-resolution (SR), aimed at reconstructing a high-resolution (HR) image from a low-resolution (LR) observation. For S-2, spectral information fusion allows for enhancing the 20- and 60-m bands to the 10-m resolution. Also, there were attempts to combine multitemporal stacks of individual S-2 bands; however, these two approaches have not been combined so far. In this article, we introduce DeepSent—a new deep network for super-resolving multitemporal series of multispectral S-2 images. It is underpinned with information fusion performed simultaneously in the spectral and temporal dimensions to generate an enlarged multispectral image (MSI). In our extensive experimental study, we demonstrate that our solution outperforms other state-of-the-art techniques that realize either multitemporal or multispectral data fusion. Furthermore, we show that the advantage of DeepSent results from how these two fusion types are combined in a single architecture, which is superior to performing such fusion in a sequential manner. Importantly, we have applied our method to super-resolve real-world S-2 images, enhancing the spatial resolution of all the spectral bands to 3.3-m nominal GSD, and we compare the outcome with very HR WorldView-2 images. We have made our implementation publicly available, and we expect it will increase the possibilities of exploiting super-resolved S-2 images in real-life applications.**

***Index Terms*— Deep learning, information fusion, multi-image super-resolution (MISR), multispectral images (MSIs), Sentinel-2 (S-2), super-resolution (SR).**

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I. Introduction

THE Sentinel-2 (S-2) mission is based on two Earth observation satellites that acquire 13-band multispectral images (MSIs) at a revisit time of around five days [1]. Free and open data policy adopted by the European Space Agency (ESA) for the S-2 products has attracted interest of the remote sensing community, leading to numerous interesting applications, including land-cover mapping [2], precision agriculture [3], water resources monitoring [4], [5], and more [6]. Spatial resolution of the individual S-2 bands ranges between 10- and 60-m ground sampling distance (GSD). The blue, green, red, and near infrared (NIR) bands ($B02$, $B03$, $B04$, and $B08$, respectively) are of 10-m GSD, the vegetation red edge bands ($B05$, $B06$, and $B07$), narrow NIR ($B08a$), and the short-wave infrared (SWIR) bands ($B11$ and $B12$) are of 20-m GSD, while the remaining coastal aerosol ($B01$), water vapor ($B09$), and cirrus clouds estimation ($B10$) bands are of 60-m GSD. While such resolution is sufficient in many cases, it also constitutes a serious limitation for the tasks that require higher accuracy in the spatial domain, such as precision farming [7] or object delineation [8], [9].

The aforementioned limitations can potentially be mitigated with effective super-resolution (SR) techniques, and therefore, the latter is attracting considerable attention from the researchers and practitioners, including many attempts to enhance spatial resolution of S-2 spectral bands [9], [10], [11], [12], [13], [14], [15], [16]. SR is a common term for a variety of techniques aimed at generating a high-resolution (HR) image from a low-resolution (LR) input, being either a single image, an image composed of multiple spectral bands, or a multitemporal series of images presenting the same scene captured at different times.

In the last decade, we have witnessed unprecedented advancements in single-image SR (SISR) attributed to the use of deep learning [17], and such techniques have been applied in a bandwise manner for enhancing S-2 images [18], [19]. However, SISR is an ill-posed problem—given a certain

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LR image, multiple possible HR solutions exist, and the larger the magnification ratio required, the more severe this problem becomes. Overall, while advanced SISR techniques lead to generating plausible images of high perceptual quality, their capabilities of reconstructing the real underlying HR information are still limited.

This problem can be addressed by using more informative input data that can be effectively fused to reconstruct the real HR information. There are two general categories of such approaches. First, SR can be based on multiple images that present the same area of interest, each of which carries a different portion of HR information. For satellite imagery, such multi-image SR (MISR) [20] is performed from images captured at different revisits of the imaging sensor, so the information fusion operates in the *temporal* domain. For that reason, this process is also termed *multitemporal* SR [21], and it has been applied for enhancing the individual S-2 bands [9], [22], [23]. Most of the state-of-the-art techniques [24], [25], [26], [27] apply the magnification factor of 3×, as they were trained from the project for on-board autonomy-vegetation (PROBA-V) dataset containing LR images of 300-m GSD resolution coupled with HR images at 100-m GSD. From the results reported in these papers, it is clear that the upsampling factor of 3× remains challenging for the state-of-the-art MISR techniques—the super-resolved images contain much more details than the LR input, but still less than the HR references. The second group of approaches, aimed at MSIs or hyperspectral images (HSIs), relies on information fusion in the *spectral* domain. It consists in exploiting spectral bands of higher spatial resolution to magnify the lower resolution bands [28]. Such techniques have been also proposed for enhancing S-2 images, and they increase resolution of the 20- and 60-m GSD bands up to 10-m GSD [11], [12], [13], [14], [29], [30].

Overall, there are two actively explored research directions for super-resolving S-2 images, namely: 1) multitemporal SR and 2) information fusion in the spectral domain. However, there were no reported attempts to combine the benefits of these two approaches, neither in a sequential manner, nor simultaneously. In this article, we address this research gap with DeepSent—a new deep network architecture for S-2 SR that fuses information from multiple spectral bands in multitemporal S-2 images. Our contribution can be summarized as follows.

1) We introduce the first SR network that benefits from information fusion performed both in the spectral and temporal domains. It combines multiple S-2 spectral bands in a multitemporal image series, enhancing the spatial resolution of the 10-, 20-, and 60-m GSD bands by a factor of 3×, 6×, and 18×, respectively. The resolution of 20- and 60-m GSD bands is first enhanced to 10-m GSD, and finally, all bands are upsampled by a factor of 3×. Importantly, DeepSent has to be trained only once to reconstruct all the bands (they are processed using the same model). The method is described in detail in Section III.
2) We report the results of our extensive experimental validation (Section IV), which indicate that the proposed simultaneous fusion in the spectral and temporal domains outperforms the state-of-the-art methods underpinned with the multitemporal image fusion [25], [26] or the spectral fusion [11]. Our ablation study clearly shows that fusion in both domains significantly improves the reconstruction accuracy for all the bands, and our simultaneous approach is more effective than combining these two fusion types in a sequential manner.
3) We demonstrate that DeepSent effectively super-resolves original S-2 data (i.e., not simulated ones like in many other approaches, as discussed in Section II-D), decreasing the nominal GSD[1] for all 12 bands down to 3.3 m (as in other works [11], [29], we do not include the $B10$ band, because it does not present any HR details). The benefits of DeepSent over the state-of-the-art techniques are confirmed with the quantitative scores obtained for the MuS2 benchmark [33] with real-world HR reference images.
4) We thoroughly investigate the reason why the artifacts are often introduced by the models trained with simulated LR data. Furthermore, we explain their origin, making it possible to identify the areas that are likely to be affected, and we present the measures that reduce the intensiveness of such artifacts.
5) The DeepSent implementation is publicly available, so that other research groups can easily reproduce the reported experiments and benefit from our approaches.

II. Related Work

We outline the state of the art in SR taking into account the category of involved information fusion, including the methods that do not exploit any fusion (Section II-A), followed by the techniques that employ fusion in the temporal domain (Section II-B) and spectral domain (Section II-C). Afterward, we present the recent advancements in real-world SR (Section II-D) to better contextualize the reported research within the efforts toward exploiting SR in practical remote sensing applications, and in Section II-E, we focus on the techniques proposed to super-resolve S-2 images.

A. Fusion-Free SR

The relation between LR and HR images can be effectively modeled using convolutional neural networks (CNNs), and the advancements in feature representation and nonlinear mapping based on deep learning are often directly exploited to improve SISR solutions [34]. They remain the most actively explored category of SR [17] and were used for enhancing the remotely sensed images, including MSIs and HSIs [35]. The first CNN proposed for SR (SRCNN) [36] was composed of just three convolutional layers, and it was shown that more complex models of much larger capacities, such as enhanced deep SR network (EDSR) [37], are more effective in modeling the LR–HR relation. Recently, it was demonstrated that SISR

[1]Nominal (or geometric) GSD indicates the pixel size, and it is not to be confused with effective (or true) GSD [31] that corresponds to the information content, also referred to as ground resolved distance [32].

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can benefit from vision transformers [38], which dynamically adjust the size of the feature maps, thus reducing the model complexity.

An important direction in SISR is concerned with the use of generative adversarial networks (GANs) [39], composed of a generator and a discriminator that compete between each other during training. Even though GANs are particularly effective in reconstructing images of high perceptual quality, they do not necessarily recover the actual HR information. Despite of that, they were explored for remote sensing [40], and in [41], it was demonstrated that imposing certain constraints on the adversarial loss may help increase the reliability of the reconstruction outcome obtained using GANs.

B. Multitemporal Image Fusion

Temporal fusion allows for higher reconstruction accuracy than SISR [20], [42], but it poses significant challenges, including image coregistration and managing temporal variability among images. Thus, the existing CNNs cannot be straightforwardly applied to MISR, and elaborating dedicated architectures has been necessary—as a result, deep learning is not that commonly exploited here as for SISR, and the techniques underpinned with conventional image processing pipelines [43] are still used for enhancing satellite images [44].

The first reported approach to exploit deep learning for MISR was the EvoNet framework [45], which employs CNNs for SISR prior to evolutionary multi-image fusion [46]. This was followed by DeepSUM—the first end-to-end deep network for MISR [21], later enhanced with nonlocal operations [47]. DeepSUM [48] assumes a fixed number of LR inputs and requires a long training, being the result of fusing the upsampled LR images. These downsides were addressed in other MISR solutions, including HighRes-net [25], that combines the latent LR representations in a recursive manner to obtain the global representation, which is upsampled to obtain the super-resolved image. Afrasiabi et al. [49] presented a thorough theoretical and experimental analysis of different MISR pipelines, which confirmed that multi-image fusion followed by upsampling allows for obtaining better reconstruction outcomes at a significantly lower computational complexity. As presented in detail in Section III, our DeepSent also adopts such approach.

In [26], the attention mechanism was exploited for selecting the most valuable features extracted from LR inputs in the residual attention multi-image SR (RAMS) network, and a recurrent network with gated recurrent units (MISR-GRU) was proposed in [50]. An et al. [24] focused on simplifying the training with the use of transformers and reported competitive results for the PROBA-V dataset [48]. Another approach is to represent an input set of LR images as a graph, which is processed with a graph neural network that produces the super-resolved image [51]. Multitemporal fusion may also be performed to combine images captured by different satellites, including S-2 and Landsat-8 [52], [53].

The existing MISR techniques can be applied to MSIs in a bandwise manner [22]. Multiple images of each band are processed in isolation to synthesize a super-resolved image of that band, but there are no reports of exploiting the spectral correlation between the bands during such temporal fusion.

C. Fusion in the Spectral Domain

Fusion in the spectral domain is primarily explored for HSIs, but the developed techniques have also been applied to MSIs, including S-2 images (discussed later in Section II-E). HSIs are often accompanied with a panchromatic channel of higher spatial resolution, which can be exploited for enhancing the remaining spectral bands. This process is known as pansharpening, and it can be effectively performed with deep learning [54]. Another possibility is to combine an LR HSI with an MSI of higher spatial resolution to obtain an HR HSI [28], [55]. Recently, it has been proposed to exploit the deep prior for regularizing the optimization unfolded into a deep network [56] that performs the MSI–HSI fusion.

When no HR channel is available, the spectral correlation between the bands can also be exploited to generate an HR image—this is commonly performed with 3-D convolutions [57] or with tensor decomposition techniques [58]. The latter may be underpinned with low-rank tensor decomposition for tensor completion [59] that is employed for recovering missing information. Such techniques were applied for HSI denoising [60], SR [61], and also for fusion of MSIs and HSIs [62]. HSI SR based on 3-D CNNs can also benefit from the attention modules that help capture local contextual features, as well as nonlocal interdependencies [63].

To ensure that the spectral information is not disturbed when an HSI is being enhanced, the spectral angle mapper (SAM) is often monitored between the reconstructed image and the ground truth [64]. Also, SAM can be embedded into the loss function during training [65] for better spectral consistency.

D. Real-World SR

Commonly, the SR algorithms are trained and validated following an artificial scenario, in which LR images are simulated by downsampling the original image, treated as an HR reference. Although the use of simulated LR images is widely adopted for SR [66], there may be a substantial performance gap between such data and real images [67], [68]. Therefore, a lot of effort is invested nowadays into creating real-life sets encompassing pairs of original LR and HR images [69] for both SISR and MISR. The PROBA-V benchmark was the first large-scale satellite image dataset [48], and recently, two datasets were published that match S-2 images with HR data. The WorldStrat dataset [70] matches S-2 with SPOT images of higher resolution, whereas the MuS2 benchmark [33] couples the 10-m S-2 bands with HR WorldView-2 images. The use of real-world datasets for evaluating SR is not straightforward, as they rely on LR and HR images captured in different conditions. Recently, Nguyen et al. [71] proposed to adapt the PROBA-V dataset for reference-aware SR, which addresses the problem of evaluation bias resulting from temporal changes among the input LR images.

The reconstruction accuracy can also be assessed in a blind way without using any reference image, but the scores may not be sensitive to the presence of artifacts [15]. Another

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approach was adopted for the OpTiGAN system [72], which was validated using images captured at Baotou Geocal site that presents different patterns at varying scale. This makes it possible to measure the effective GSD from a super-resolved image.

An important research direction that attracts increasing attention is concerned with evaluating SR in the context of specific image analysis tasks. Xu et al. [73] demonstrated the benefits of employing SISR for S-2 images to improve the quality of their semantic segmentation system, applied for detecting building structures. In [9], the MISR HighRes-net network was validated for a similar task with the ground-truth data derived from the SpaceNet-7 dataset [74]—for every image, a multitemporal series of S-2 images was collected and super-resolved before performing building delineation. In [75], an existing SISR technique [76] was exploited for change detection performed from prechange and postchange images of different spatial resolutions—an image of a larger GSD is super-resolved before being compared with an image at a finer scale. Recently, we have evaluated various HSI SR techniques in specific test cases concerned with precision farming, air pollution, and water quality estimation [77]. Overall, such task-based evaluation is an interesting emerging direction of real-world SR, but no benchmarks have been established for S-2 images so far, and the reference-based assessment remains a commonly adopted standard procedure.

E. Super-Resolving S-2 Images

S-2 SR has already attracted considerable attention. In [10], SRCNN [36] was exploited for super-resolving the RGB S-2 bands (i.e., $B02$, $B03$, and $B04$). These bands were processed independently, and more advanced SISR techniques, including EDSR [37], were also applied in a bandwise manner to enhance original multispectral S-2 images [19] (this method was evaluated quantitatively using 2.5-m PlanetScope images). GANs were also used for super-resolving S-2 images in a bandwise manner [78]. Beaulieu et al. [79] employed them for upsampling the 10-m bands to 2.5 m, and Salgueiro Romero et al. [18] used enhanced super-resolution GAN (ESRGAN) [80] for enhancing the original S-2 images by a factor of 5×. Importantly, it was noticed in [18] that the visual quality is often obtained by hallucinating image details, hence subject to a trade-off with the quantitative metrics, which indicate the accuracy of reconstructing the actual HR information. In [18] and [79], the HR WorldView images were exploited for quantitative evaluation. Recently, Zabalza and Bernardini [81] trained a residual network with spectral attention using real-world pairs of S-2 10-m bands coupled with HR PlanetScope data.

Also, there were a few attempts to enhance multitemporal S-2 images in a bandwise manner. Vaqueiro et al. [23] proposed to exploit the georeferencing error when fusing S-2 images using the k-nearest pixels technique. In [27], enforcing permutation invariance within a set of LR inputs was shown to improve the results for the PROBA-V images and S-2 $B03$ band with simulated LR images. Also, RAMS [26] and HighRes-net [25] trained with real-world PROBA-V images were used for enhancing multitemporal series of individual S-2 bands, and Razzak et al. [9] enriched HighRes-net with a radiometric consistency loss before applying it for MISR of S-2 images.

A widely explored direction is concerned with exploiting the 10-m S-2 bands to super-resolve the 20- and 60-m bands relying on spectral fusion [82]. Brodu [83] adapted the existing pansharpening techniques to super-resolve the 20- and 60-m S-2 bands using a simulated panchromatic channel obtained from the 10-m RGB bands. Lanaras et al. [84] analyzed correlations between the spectral bands and exploited the textural information from higher resolution bands to magnify the remaining ones. Their solution was followed with DSen2—a deep CNN architecture elaborated to magnify all the bands to 10-m GSD [11]. As in other works [29], the $B10$ band was not included. A similar fusion for super-resolving the $B11$ SWIR band was reported by Gargiulo et al. [85], and it was extended to process all of the 20-m bands [13]. The SR outcome was inspected qualitatively against the artifacts and quantitatively against spectral distortions in a reference-free manner. Paris et al. [12] applied their S-2 sharpening method based on 3-D filtering for real-world images and inspected the results qualitatively. In [14], the reduced-rank approximation was employed for S-2 sharpening, and in [16], the deep image prior was employed. Armannsson et al. [30] compared several S-2 sharpening approaches and applied the Bayesian optimization to tune their hyperparameters. All of these methods are limited to enhancing the resolution of LR bands to 10-m nominal GSD, but S-2 images can also be fused with HR imagery. In [86], PlanetScope images were used to sharpen the 10- and 20-m S-2 bands to 2.5 m.

Lloyd et al. [87] proposed to extract HR information from S-2 optical bands and fuse it with Sentinel-3 thermal bands for measuring sea surface temperature. Lin and Bioucas-Dias [29] introduced a self-similarity prior that is used as a regularizer when super-resolving the lower resolution S-2 bands. Their method, in contrast to the techniques mentioned earlier in this section, was fed with both simulated and original S-2 data, and the obtained results were inspected qualitatively.

Overall, there are many S-2 SR techniques underpinned with information fusion realized in the temporal or spectral dimension, but there were no attempts reported yet to combine multitemporal and multispectral information at once. Also, just a few methods were validated quantitatively using real-world HR images. Some researchers observed artifacts when the models trained using simulated data were fed with real-world images [13], [18], [22], but the attempts to understand their origin and reduce their intensiveness were rather limited, and this problem is also tackled in the study reported here.

III. Method

In this section, we present DeepSent[2]—a fully convolutional deep neural network for super-resolving S-2 images by means of spectral (multiple bands) and temporal (multiple images per band) information fusion. Such an approach allows us to upsample all the bands up to 3.3-m GSD resolution; hence, the magnification factors are of 3×, 6×, and 18× for 10-, 20-, and 60-m GSD bands, respectively. As discussed in

[2]Implementation of DeepSent is available at https://gitlab.com/tarasiewicztomasz/deepsent upon acceptance.

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Section III-A, the resolution of the learned representation of the 60-m GSD bands is first raised by a factor of 3× to the resolution of the 20-m bands, and their common representation is upsampled by a factor of 2× and fused with the 10-m GSD bands. Finally, the obtained feature maps are upsampled by a factor of 3×, as in many MISR techniques underpinned with deep CNNs [24], [25], [26], [27]. We can distinguish three main parts of the architecture: the band fusion (Section III-B), cross-resolution fusion (Section III-C), and super-resolution (Section III-D) modules.

A. Overview of the Architecture

Our DeepSent architecture that benefits from the spectro-temporal information fusion is presented in Fig. 1. As recommended in [49], we first fuse the images in the temporal and spectral domains, and then, the obtained representation is upsampled to the required size. A temporal series of $N_{\mathcal{I}}$ MSIs ($\mathcal{I}$) is organized into 12 stacks, each of which contains $N_{\mathcal{I}}$ images ($\mathcal{I}^b$) of the same band b. In the *band fusion module* (Section III-B), the input bands ($\boldsymbol{B}_{\text{all}}$) are grouped by their spatial resolution; hence, we obtain three mutually exclusive S-2 band subsets: $\boldsymbol{B}_{10\,\text{m}} = \{B02, B03, B04, B08\}$, $\boldsymbol{B}_{20\,\text{m}} = \{B05, B06, B07, B08\text{a}\}$, and $\boldsymbol{B}_{60\,\text{m}} = \{B01, B09\}$ with 10-, 20-, and 60-m bands, respectively. We utilize the S-2 Level-2A products, which contain 12 orthorectified atmospherically corrected surface reflectance bands—the $B10$ band is excluded, as it is an uncalibrated band exploited for detecting thin cirrus clouds (it is referred to as the "cirrus" band) [88].

Having the bands grouped, we encode LR images and fuse the temporal information for each band independently. After that, we combine the spectral information between the bands assigned to the same spatial resolution group $\boldsymbol{B}_r$, where $r \in \{10, 20, 60\ \text{m}\}$, resulting in three stacks of feature maps, each of different spatial resolutions. Afterward, we upsample and merge them sequentially to reach the final latent representation of a scene at 10-m GSD using the *cross-resolution fusion module* (Section III-C). Therefore, 60-m feature maps are upsampled 3× to match the resolution of 20-m GSD, and then, we recursively combine them with the $\boldsymbol{B}_{20\,\text{m}}$ bands. We apply the same procedure to match the result of this operation with 10-m bands through upsampling them 2×. Such representation of a scene is decoded and upsampled 3× to reach the target resolution of 3.3-m GSD using the *SR module* (Section III-D). The intrinsic characteristics of the band being reconstructed are captured and propagated to the output using a skip connection that transfers the averaged stack of bicubically upsampled input images for that band. It is ultimately concatenated with the decoded scene and merged using a single convolutional layer. This is repeated for every band to reconstruct the entire MSI.

B. Band Fusion Module

We exploit a recursive fusion block inspired by the HighRes-net architecture [25] to perform fusion in the temporal dimension, followed by fusion in the spectral dimension. Recursive fusion was originally applied for merging a single stack of multitemporal LR images—here, we exploit it for this purpose (the first recursive fusion blocks at each bandwise processing path) as well as to fuse the features extracted from the individual spectral bands (the recursive fusion blocks later on the processing paths). Here, we extend this idea and integrate feature maps obtained for different input bands of the same spatial resolution. Since the characteristics of the available S-2 bands can significantly differ, all input images undergo z-score normalization, performed for each band independently. Afterward, we exploit the embedding blocks EB_b and apply them to each band of an input LR image. The EBs consist of the convolutional layers and residual blocks followed by the parametric rectified linear unit (PReLU) activation, and they extract 64 feature maps for each $\mathcal{I}^b$. In addition, an EB implicitly coregisters the input images (for each band separately), as each one is collated with the median band m, in which each pixel at the (x, y) position is the median pixel value from the entire LR stack. Thus, the embedded outputs become

$$\mathbf{E}_b = \left[\mathbf{e}_1^b, \mathbf{e}_2^b, \ldots, \mathbf{e}_{N_{\mathcal{I}}}^b\right] \tag{1}$$

where

$$\mathbf{e}_i^b = \text{EB}_b\left(\mathcal{I}_i^b, m_b\right) \in \mathbb{R}^{64\times H_r \times W_r} \tag{2}$$

and $\mathcal{I}_i^b$ denotes the bth band of the ith LR image, m_b is the median bth band, and H_r and W_r are the height and width of the LR image of spatial resolution r, respectively.

The temporal information fusion is achieved using the recursive approach, which extracts the mutual features from the stack of input images and halves their number with each iteration until all the images are fused into one. If $N_{\mathcal{I}}$ is not the power of 2, we pad the embedded tensor $\mathbf{E}_b$ with zeros and reshape it to $(N'_{\mathcal{I}}, 64, H_r, W_r)$, where $N'_{\mathcal{I}} \in \{1, 2, 4, 8, \ldots\}$, and such conditionally padded tensor is $\mathbf{E'}_b$. In addition, we create a vector $\alpha_b^0 \in \mathbb{B}^{N'_{\mathcal{I}}}$ indicating whether a specific temporal image in $\mathbf{E}'$ was padded (zero) or not (one). The recursive fusion block FB for each band $b \in B$ operates on the corresponding $\mathbf{E'}_b$, and it iteratively halves its temporal dimension until it reaches 1. We merge the information transferred directly from EBs—in each fusing iteration $k \in \{0, 1, \ldots, K-1\}$, where $K = \log_2(N'_{\mathcal{I}})$, we generate the fusion map

$$X_b^{k+1} = s_1^k + \alpha_2^k f_b\left(s_1^k, s_2^k\right) \in \mathbb{R}^{N^{k+1}\times 64\times H_r \times W_r} \tag{3}$$

where $N^{k+1} = N^k/2$, $\alpha_b^{k+1} = \alpha_1^k \in \mathbb{B}^{N^{k+1}}$, f_b is a band-specific residual block built with a skip connection and two convolutional layers, and s_1^k and s_2^k represent two halves of X_b^k split along its temporal dimension. Similarly, we divide the α_b^k in two equisized vectors, where α_1^k denotes its first half and α_2^k becomes the second one. Also, to initiate the fusion loop, we set $X_b^0 = \mathbf{E'}_b$ and $N^0 = N'_{\mathcal{I}}$. Therefore, the output of the fusion block FB_b can be defined as follows:

$$X_b^K = \text{FB}_b\left(\mathbf{E'}_b, \alpha_b^0\right) \in \mathbb{R}^{64\times H_r \times W_r}. \tag{4}$$

The spectral fusion uses an analogous recursive method as the temporal one, but instead of combining a stack of LR images from a single band, we concatenate and merge

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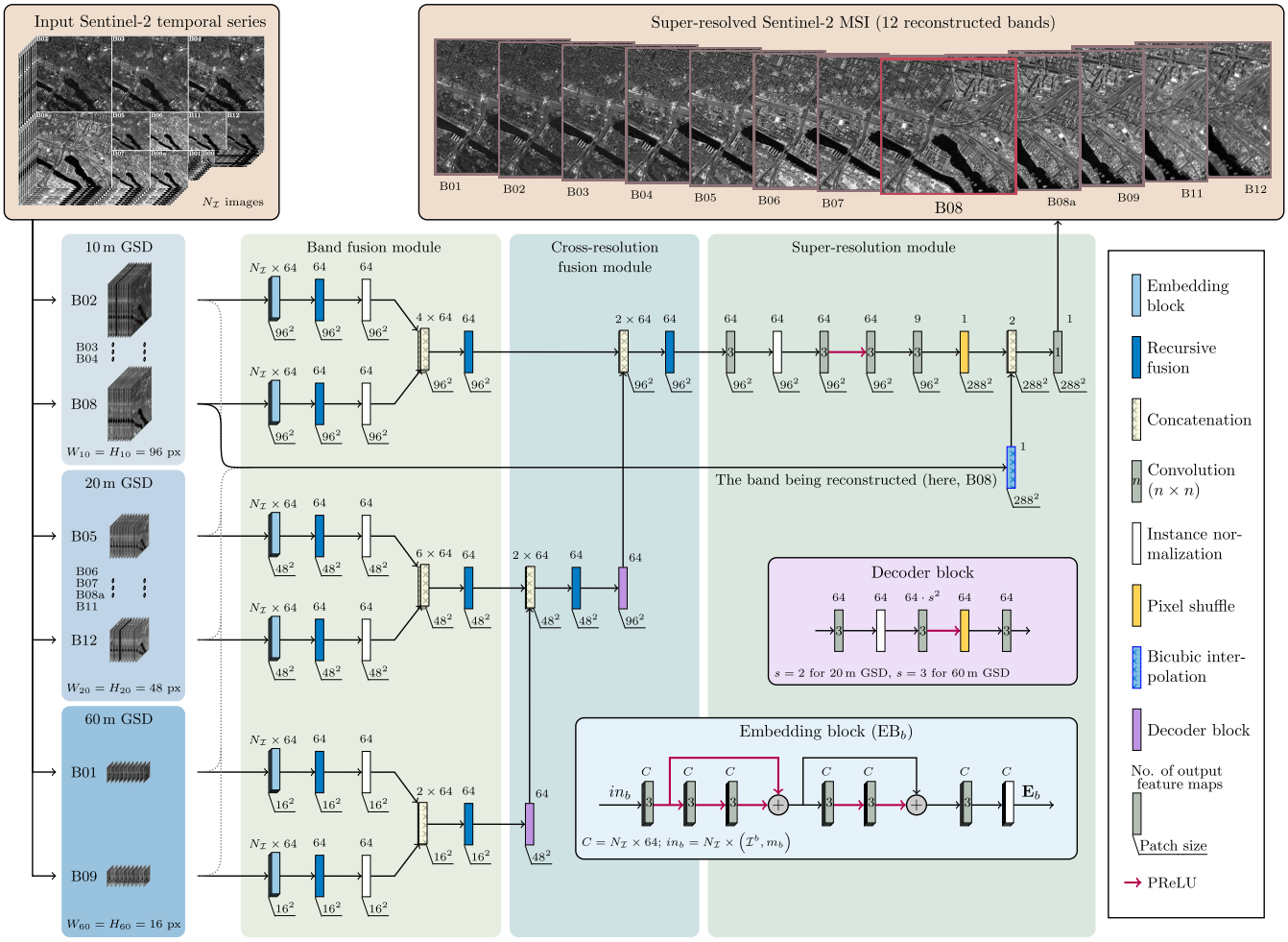

Fig. 1. DeepSent architecture. A stack of input S-2 MSIs is processed, and every band of the output super-resolved image is reconstructed by passing the appropriate stack of the input bands—in the figure, this is presented for the $B08$ band, and the remaining alternative connections are shown with gray dotted lines.

fused features of bands having the same spatial resolution. It allows for an additional flow of information and enhancing the scene representation in the latent space. We apply the instance normalization layer for every fused feature map X_b^k to minimize the luminance differences between the bands, and focus more on their textural features. Then, we concatenate the feature maps of bands belonging to the same subset $\boldsymbol{B}_r$ and pass them to the recursive fusion layers, defined exclusively for each $\boldsymbol{B}_r$. The only conceptual difference is that the first dimension represents the spectral entries and not the temporal ones. Similar to the temporal fusion, we also pad the input tensors to match their size to a power of 2 and provide the vectors indicating which entry was padded. Such recursion performed on multiple levels ensures flexibility of our model in terms of the input data, as it does not require passing the same number of LR images for every band. Moreover, it can super-resolve a scene even if we provide LR images only for some selected bands—it will be experimentally shown in Section IV.

C. Cross-Resolution Fusion Module

To combine information extracted from bands of different resolutions, we sequentially upsample the feature maps to the 10-m GSD resolution. We perform the upsampling in multiple parts of the network using pixel shuffling [26]. It reorganizes elements in the input tensor with the dimensionality of $(Cs^2 \times H_T \times W_T)$ to $(C \times sH_T \times sW_T)$, where C represents the number of input channels and s denotes the upsampling factor. In the upsampling block, we utilize a 2-D convolution (with the instance normalization), followed by another 2-D convolution, which becomes the input to the pixel shuffle operation. Finally, its output is passed to the last convolutional layer, which prunes possible artifacts from the upsampled image. Importantly, the second convolutional layer multiplies the number of feature maps by a square of the block's upsampling factor. The reason for that is to ensure that the upsampled image consists of the same number of feature maps as the input one.

Afterward, the 60-m latent scene representation is upsampled by a factor of 3, fused with the 20-m feature maps using concatenation followed by a recursive fusion layer, and then fed into the upsampling layer of the 20-m processing path. This is later merged with the 10-m bands in the same way, as the 60-to-20-m fusion is performed. Here, we use the recursive fusion layer that was previously exploited to combine the temporal and spectral features. It allows us to make DeepSent functional over incomplete input data—in this case, it can perform a successful forward pass even if the

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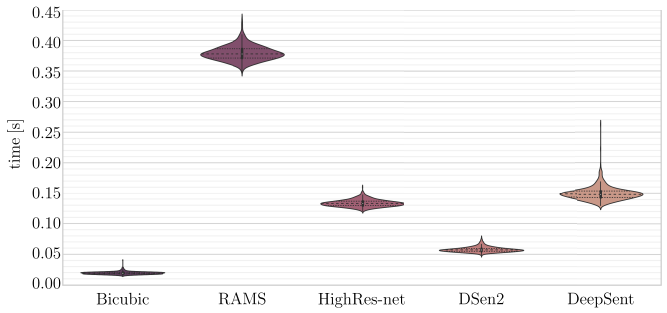

Fig. 2. Distribution of the processing times necessary to reconstruct an MSI of 288 × 288 pixels. The dashed lines indicate the median and quartile values.

bands of a given resolution are missing in the input LR image stack.

D. SR Module

At this point, the model stores multiple embedded feature maps at 10-m GSD resulted from the fusion performed on multiple levels of the DeepSent architecture. To upsample the images even further (to the target 3.3-m GSD), we transform the embedded scene representation using pixel shuffling. This operation requires the number of channels in an input image to be divisible by a square of the upsampling factor. Thus, we pass it through a convolutional layer, which generates nine feature maps with subpixel shifts between them, and then, the pixel shuffle transforms these maps into a single-channel tensor with the tripled spatial dimensions. We utilize 12 such convolutional layers (one for each S-2 band) and exploit them as a reconstruction mechanism, which recreates the band-specific characteristics based on the latent representation.

Since the original information captured by each S-2 band is entangled in the latent space, it may not be straightforward to reconstruct its specific spectral characteristics. To address this issue, we benefit from the skip connections, which bypass the input bands of interest directly to the SR module, averaged across the temporal dimension and bicubically upsampled to match the output resolution of the model. Finally, as DeepSent operates on standardized images, the resulting super-resolved images are standardized as well.

IV. Experimental Validation

In this section, we present and discuss the results of our experimental study performed over both simulated and real-life image data, including the recently published MuS2 benchmark [33]. The objectives of the experiments are twofold: 1) to investigate the reconstruction capabilities of DeepSent and to confront it with the existing SR methods benefiting from the temporal or spectral information fusion for both simulated and real-world images and 2) to thoroughly understand the benefits of spectral and temporal fusion and to justify the proposed processing flow.

A. Experimental Setup

In the reported study, we have used the following datasets: 1) simulated data obtained from original S-2 images (termed Sim-S2); 2) data simulated from HR HSIs (termed Sim-HSI); 3) real-world S-2 images without HR reference (termed Org-S2); and 4) the MuS2 benchmark [33] with real-world S-2 images, whose 10-m bands are matched with corresponding HR WorldView-2 bands (downsampled to 3.3-m GSD). When HR reference images are available, we assess the reconstruction quality using peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), learned perceptual image patch similarity (LPIPS) [89], and SAM (the better reconstruction is indicated with higher PSNR and SSIM, and lower LPIPS and SAM scores). For assessing MISR, the PSNR and SSIM metrics require compensation with regards to the average brightness and full-pixel displacements between SR output and its corresponding HR reference. Therefore, as proposed

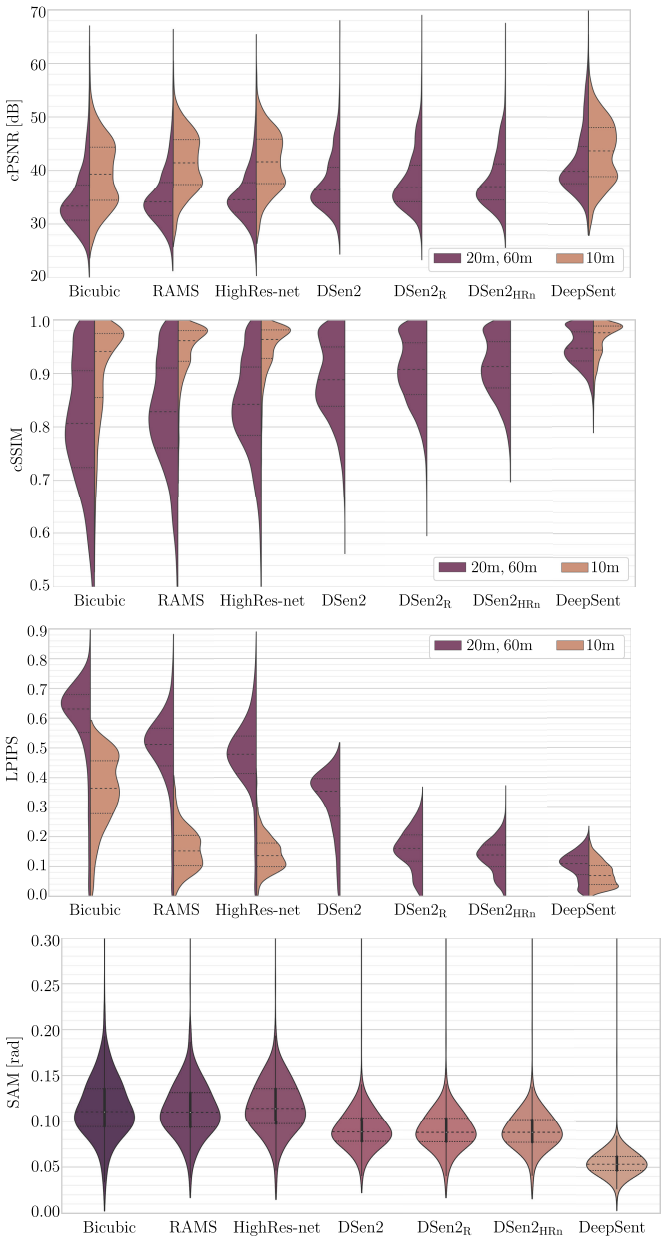

Fig. 3. Distribution of the reconstruction accuracy scores obtained for Sim-S2 dataset (its test set part), presented for the bands grouped according to their resolution. The dashed lines indicate the median and quartile values.

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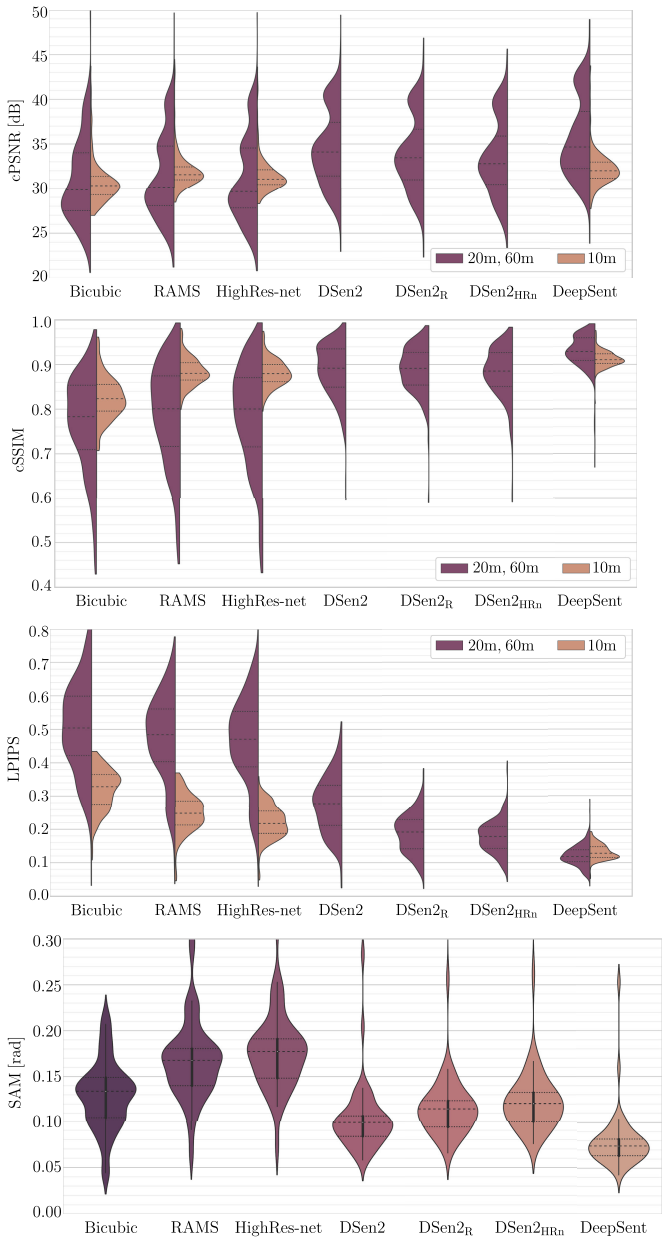

Fig. 4. Distribution of the reconstruction accuracy scores obtained for the Sim-HSI images, presented for the bands grouped according to their resolution. The dashed lines indicate the median and quartile values.

in [48] and followed in many works on MISR [21], [25], [26], we report the scores after applying such compensation, and these adjusted metrics are termed cPSNR and cSSIM. For the Org-S2 images, which are not coupled with any HR reference, we compute SAM after downsampling the super-resolved image back to the LR size to verify the radiometric consistency, and the quality of the super-resolved image is assessed using the reference-less naturalness image quality evaluator (NIQE) [90]. To verify whether the differences between the scores obtained using DeepSent and other methods are statistically significant, we employed the two-tailed Wilcoxon signed-rank test (at $p = 0.05$) with the null hypothesis saying that two different methods allow for the same reconstruction quality. The differences between two techniques are considered significant, if the null hypothesis can be rejected at $p < 0.05$.

In order to create the Sim-S2 and Org-S2 datasets, we collected eight different scenes with a total coverage of around 100 000 km^2, each composed of 15 geographically coregistered S-2 Level-2A tiles captured at different times. Each tile comprises cloud masks and 12 square-shaped images (10 980 $\times$ 10 980, 5490 $\times$ 5490, and 1830 $\times$ 1830 pixels for 10-, 20-, and 60-m GSD bands, respectively). To generate the Sim-S2 dataset, we first downsampled the 10-m bands in the original S-2 images 6$\times$ and 20-m bands 3$\times$, so that the resolution of all bands is uniform and equal to the original resolution of the 60-m bands. This was performed to prepare an HR reference for DeepSent, as it super-resolves all the bands to the same resolution. Next, the tiles were split into 288 $\times$ 288 patches (each composed of 12 bands), and we simulated nine LR MSIs from each patch (later treated as our target HR). The simulation consisted of the following steps: 1) translating a patch with a random subpixel shift; 2) manipulating its contrast and brightness; and 3) contaminating it with Gaussian blur and additive Gaussian noise. These operations were performed independently for each band within an image, except for the translations, which were shared across the bands—this was to simulate different spectral angles between the input LR images, because they are also observed for original S-2 data, as explained later in Section II-D. Finally, to convert such patches into LR input data, we downsampled them bicubically by the factors of 3$\times$ (to simulate the 10-m bands), 6$\times$ (to get the 20-m bands), and 18$\times$ (to get the 60-m bands). Therefore, we obtained the square images of 96 $\times$ 96, 48 $\times$ 48, and 16 $\times$ 16 pixels for 10-, 20-, and 60-m bands, respectively. Overall, we obtained 4320 multispectral patches that have been split into training (80% of patches), validation (10%), and test sets (10%), without any overlaps in the spatial domain. The training set retrieved from the Sim-S2 images was used to train DeepSent and other techniques considered in this study.

To create the Org-S2 dataset (without HR references), the original Level-2A S-2 tiles were partitioned into 288 $\times$ 288 patches for the 10-m bands, 144 $\times$ 144 for 20-m bands, and 48 $\times$ 48 for the 60-m bands, and we picked the patches located inside the patches selected earlier to the Sim-S2 test set. A single patch in Sim-S2 covers the same region (at a different scale) as 36 patches from the Org-S2 set. For each of these patches, we have 15 MSIs acquired at different times, out of which we selected nine with the smallest cloud coverage (based on the cloud masks). Therefore, our Org-S2 set is composed of 1044 MSI stacks, each one encompassing nine observations acquired at different times (a single observation being a cropped Level-2A S-2 MSI at its original resolution).

The Sim-HSI dataset was generated using the hyperspectral data acquired using Airborne Visible/Infrared Imaging Spectrometer (AVIRIS)[3] at 3.4-m resolution. We extracted 42 patches of 486 $\times$ 486 pixels each from the f130818t01p00r08 flight presenting urban area and different types of terrain. At first, we simulated the multispectral bands

[3]The AVIRIS data are available at https://aviris.jpl.nasa.gov/dataportal.

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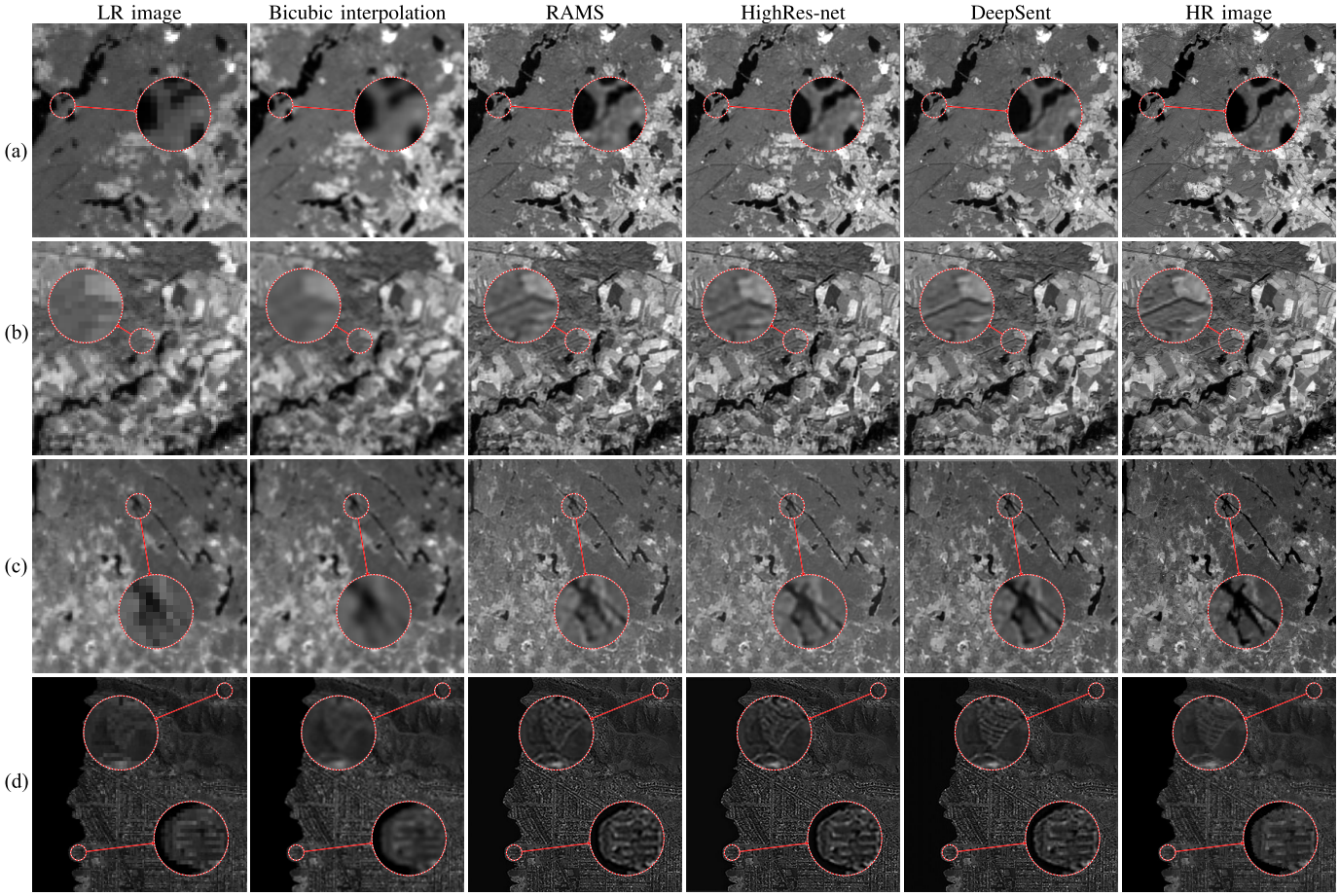

Fig. 5. Super-resolved band $B08$ images [from (a)–(c) Sim-S2and (d) Sim-HSIdatasets] obtained using different techniques. The reconstruction was performed from multiple simulated LR images—one of them is shown for each region along with the HR reference. For Sim-S2 images, the HR images are of 60-m GSD, and for the Sim-HSI images, the HR reference is of 3.4-m GSD. The results for the remaining bands are included in the Supplementary Material.

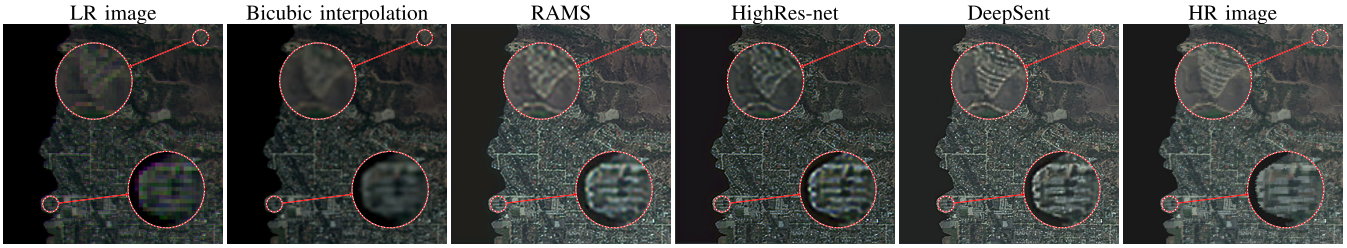

Fig. 6. Color image composed from 10-m $B02$, $B03$, and $B04$ bands for the image from the Sim-HSI dataset, reconstructed using different techniques. The reconstruction was performed from multiple simulated LR images—one of them is shown for each region along with the HR reference. The results for the remaining bands are included in the Supplementary Material.

of the same characteristic as that of S-2 MSIs [91], [92]. They were treated as the HR reference, and the LR data were simulated following the same protocol as for the Sim-S2 dataset, resulting in nine LR input images sized 162 × 162, 81 × 81, and 27 × 27 pixels for the simulated 10-, 20-, and 60-m bands, respectively, for each HR patch. While Sim-S2 has been obtained from real S-2 images (preserving the original spectral properties of S-2 images), the Sim-HSI is supposed to better reflect the scale, as the spatial resolution of the input HSI equals 3.4-m GSD; thus, the simulated LR images are at the same scale as original S-2 images.

DeepSent is confronted with two state-of-the-art MISR architectures, RAMS [26] and HighRes-net [25], that enhance the spatial resolution 3×, as well as with the DSen2 [11] network that operates from a single MSI and enhances the resolution of 20- and 60-m bands up to 10-m GSD. To make our experiments more comprehensive, we have combined DSen2 with RAMS and HighRes-net in a sequential manner, so that the output of the bandwise multitemporal SR (performed with the latter models) is directly transferred to DSen2 (thus, it operates on 60- and 20-m bands already enlarged 3×). We denote such conjunctions as DSen2_{R} and $\text{DSen2}_{\text{HRn}}$ for RAMS and HighRes-net, respectively. Such combinations have the same upsampling capabilities as DeepSent, and similar to our method, they exploit fusion of both temporal and spectral information. As, in some cases, the images produced by other

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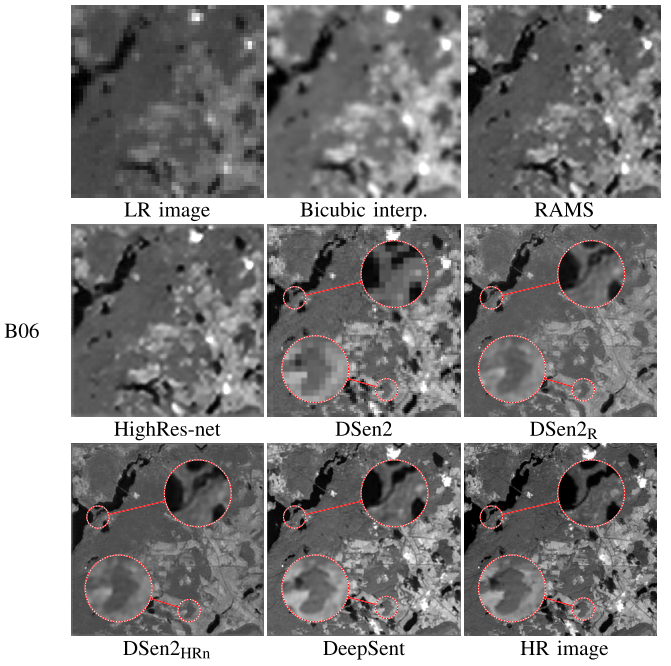

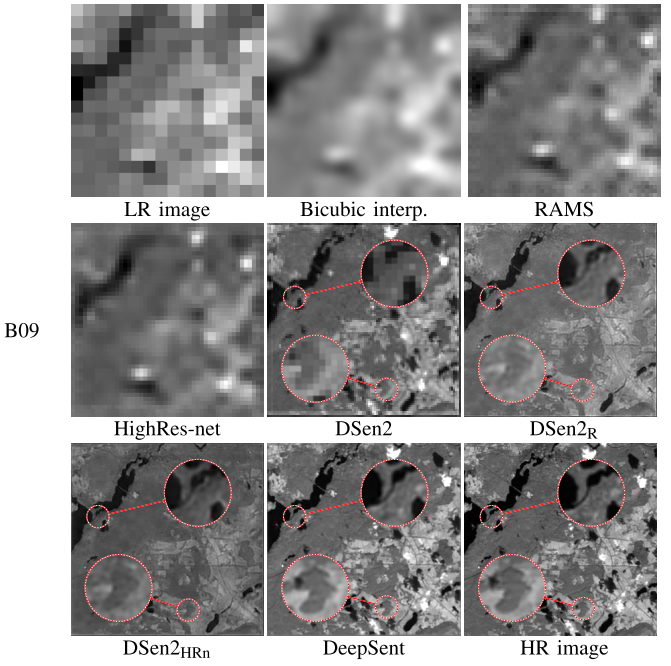

Fig. 7. Super-resolved images obtained by different methods for the *B*06 (20-m GSD) and *B*09 (60-m GSD) bands. The reconstruction was performed from multiple simulated LR images—one of them is shown along with the HR reference. The results for the remaining bands are included in the Supplementary Material.

methods are of lower spatial resolution than the size of the HR reference that matches the upsampling capabilities of DeepSent (3× larger than the input 10-m bands), we bicubically upsample these images to compare them with the reference ones. The images (for all the bands) generated by DeepSent, $\text{DSen2}_{\text{HRn}}$, DSen2_{R}, and the super-resolved images of 10-m bands for RAMS and HighRes-net are of the same spatial resolution, and they do not require any additional adjustments.

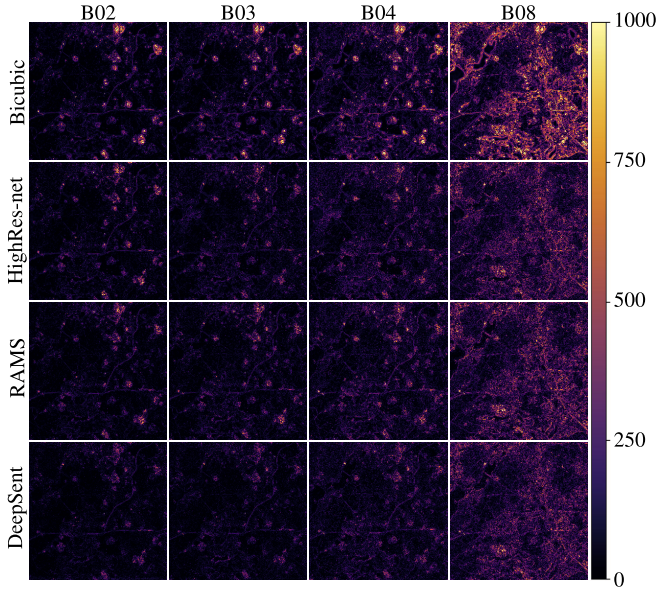

Fig. 8. Absolute differences between the super-resolved images (for the same scene as in Figs. 5(a) and 7) and their HR counterparts for 10-m bands at the 14-bit depth.

We have trained RAMS, HighRes-net, and DSen2 using the LR images from the Sim-S2 set, and we bicubically readjusted the size of all HR target images to match the shape of their output. Thus, HR images used to train HighRes-net and RAMS networks are 3× larger than their LR counterparts, while for DSen2, we maintain the upsampling factor of 6× and 2× for 60- and 20-m bands. Moreover, we ensured that each model was provided with the training samples in the very same order during training. The batch for DeepSent includes four patches, and we utilized the Adam optimizer with the learning rate of $2 \cdot 10^{-4}$, maximum of 200 epochs, and the early stopping condition was activated, if there was no improvement in the loss value in 15 consecutive epochs. As the loss function, we utilized the corrected mean-squared error (cMSE), which is widely applied for MISR problems [50], [93] and tightly related to the cPSNR metric. The training hyperparameters (excluding the batch size, as discussed below) were kept consistent across all investigated architectures. For HighRes-net and RAMS, we used the batches of 32, 64, and 196 patches for 10-, 20-, and 60-m S-2 bands, and we trained the DSen2 models for reconstructing 20- and 60-m bands with the batch sizes of 24 and 16 patches. This is strictly related to the available memory of our GPU device, which is NVIDIA RTX 3090 equipped with 24-GB VRAM (the number of learnable parameters in DeepSent is around 9.2×10^6). Our techniques were implemented in Python 3.7 and PyTorch 1.10 deep learning library. The distribution of processing times needed to reconstruct a 12-band multispectral patch of 288 × 288 pixels is presented in Fig. 2 (DSen2 reconstructs eight bands, as it cannot be applied to the 10-m bands). DeepSent is slightly (though significantly in the statistical sense) slower than HighRes-net, but over 2× faster than RAMS.

B. Validation Results for the Simulated Data

In Table I, we gather the metrics for every model over the simulated data (test set for Sim-S2 and Sim-HSI), and

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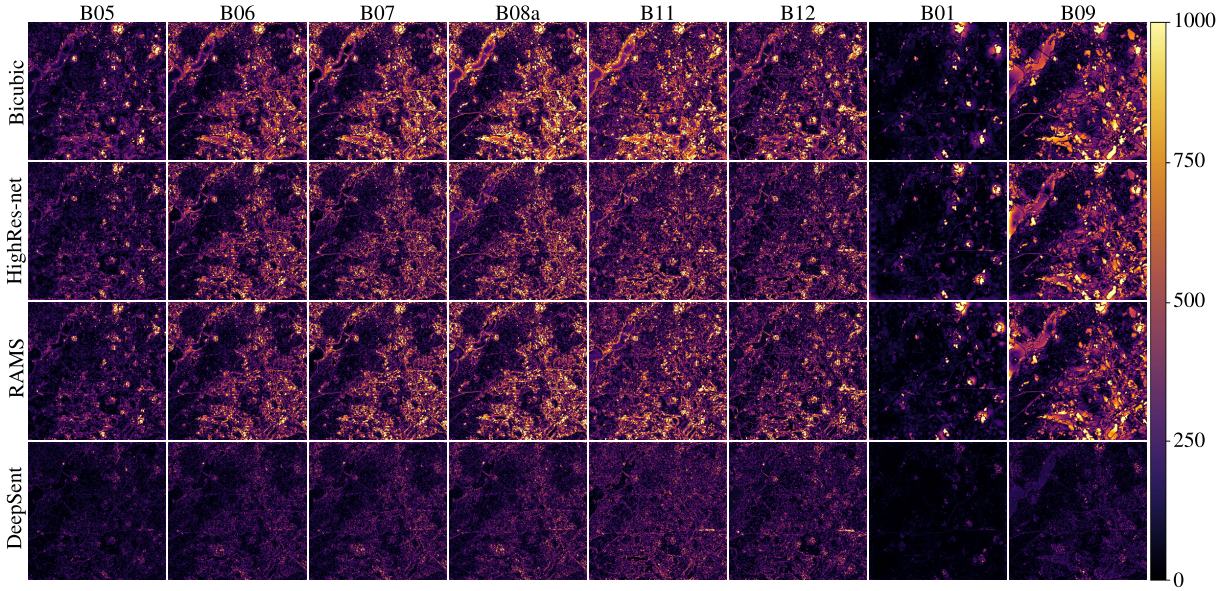

Fig. 9. Absolute differences between the super-resolved images (for the same scene as in Figs. 5(a) and 7) and their HR counterparts for 20- and 60-m bands at the 14-bit depth.

TABLE I

RECONSTRUCTION ACCURACY SCORES OBTAINED FOR THE SIMULATED DATA (SIM-S2 AND SIM-HSI), CALCULATED FOR EVERY BAND, AND AVERAGED WITHIN EACH SPATIAL RESOLUTION GROUP FOR THE IMAGES IN THE TEST SET. THE BEST RESULTS IN EACH GROUP ARE BOLDFACED, AND THE SECOND BEST ARE UNDERLINED

Simulated dataset	Method	Metric→ Bands →	cPSNR 60 m	cPSNR 20 m	cPSNR 10 m	cSSIM 60 m	cSSIM 20 m	cSSIM 10 m	LPIPS 60 m	LPIPS 20 m	LPIPS 10 m	SAM
Sim-S2 (test set)	Bicubic		38.66	33.07	39.46	0.9040	0.7697	0.9066	0.4564	0.6228	0.3547	0.1155
	RAMS		38.26	33.85	41.50	0.9036	0.7986	0.9457	0.4447	0.4978	0.1571	0.1130
	HighRes-net		38.11	34.25	41.55	0.9035	0.8180	0.9487	0.4465	0.4706	0.1425	0.1171
	DSen2		42.74	35.07	—	0.9602	0.8485	—	0.2034	0.3617	—	0.0915
	$DSen2_R$		43.68	36.16	—	0.9702	0.8820	—	0.0868	0.1831	—	0.0909
	$DSen2_{HRn}$		43.60	36.34	—	0.9720	0.8912	—	0.0728	0.1550	—	0.0903
	DeepSent		**49.06**	**39.26**	**43.68**	**0.9872**	**0.9348**	**0.9629**	**0.0450**	**0.1219**	**0.0735**	**0.0553**
Sim-HSI	Bicubic		32.29	30.78	30.70	0.8077	0.7565	0.8253	0.6162	0.4682	0.3183	0.1293
	RAMS		32.18	31.21	32.08	0.8399	0.7717	0.8844	0.5746	0.4415	0.2488	0.1677
	HighRes-net		31.70	30.92	31.53	0.8338	0.7684	0.8817	0.5737	0.4249	0.2215	0.1815
	DSen2		37.19	33.57	—	0.9327	0.8717	—	0.2177	0.2936	—	0.1031
	$DSen2_R$		36.63	33.05	—	0.9294	0.8744	—	0.1745	0.1936	—	0.1205
	$DSen2_{HRn}$		35.94	32.57	—	0.9244	0.8672	—	0.1538	0.1855	—	0.1282
	DeepSent		**38.59**	**34.73**	**32.28**	**0.9566**	**0.9186**	**0.9124**	**0.1065**	**0.1260**	**0.1322**	**0.0793**

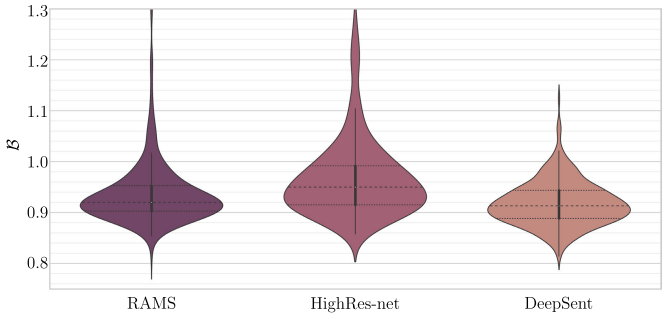

Fig. 10. Distribution of the balanced score ($\mathcal{B}$) obtained for the MuS2 benchmark dataset. The dashed lines indicate the median and quartile values.

in Figs. 3 and 4, we show the distribution of the scores for the bands, grouped by their GSD resolution.[4] DSen2 effectively benefits from the information contained in the higher resolution spectral bands when reconstructing the 20- and 60-m bands, and it surpasses both RAMS and HighRes-net. This includes the 20-m bands, even though DSen2 upsamples them 2×, compared with 3× for RAMS and HighRes-net—this shows the benefits of exploiting information from the 10-m bands. Although multitemporal data fusion is affected by the natural variability of the Earth surface appearance, primarily caused by the vegetation characteristics, RAMS and HighRes-net offer substantial gain over the bicubic interpolation for 20- and 10-m bands, similar to that reported for the PROBA-V dataset [25], [26]. For the Sim-S2 dataset, the $DSen2_R$ and $DSen2_{HRn}$ variants that combine the multispectral and multitemporal data fusion in a sequential manner render higher scores (with all metrics) than any of their components standalone, and for Sim-HSI some of the metrics, including SAM, indicate DSen2 as a more accurate technique. However, in all the cases, it is DeepSent, which delivers significantly better results than all other techniques (in the statistical sense, verified for all the metrics, and in all bands separately).

[4]The detailed quantitative scores for every band are presented in the Supplementary Material available at https://gitlab.com/tarasiewicztomasz/deepsent.

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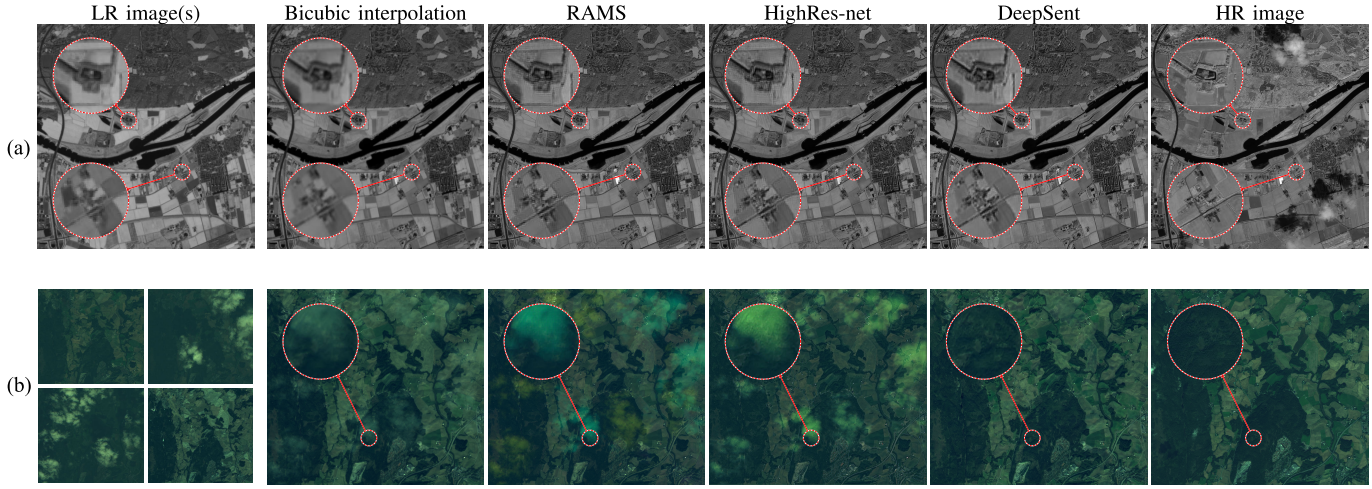

Fig. 11. Example of super-resolving images from the MuS2 dataset using different techniques—(a) $B08$ band, and a color image composed of (b) $B02$, $B03$, and $B04$ bands. The reconstruction was performed from multiple simulated LR images—for (a), one of them is shown, and for (b), four examples are presented to show the differences between them. Also, the HR reference acquired with WorldView-2 is provided. The results for the remaining bands are included in the Supplementary Material.

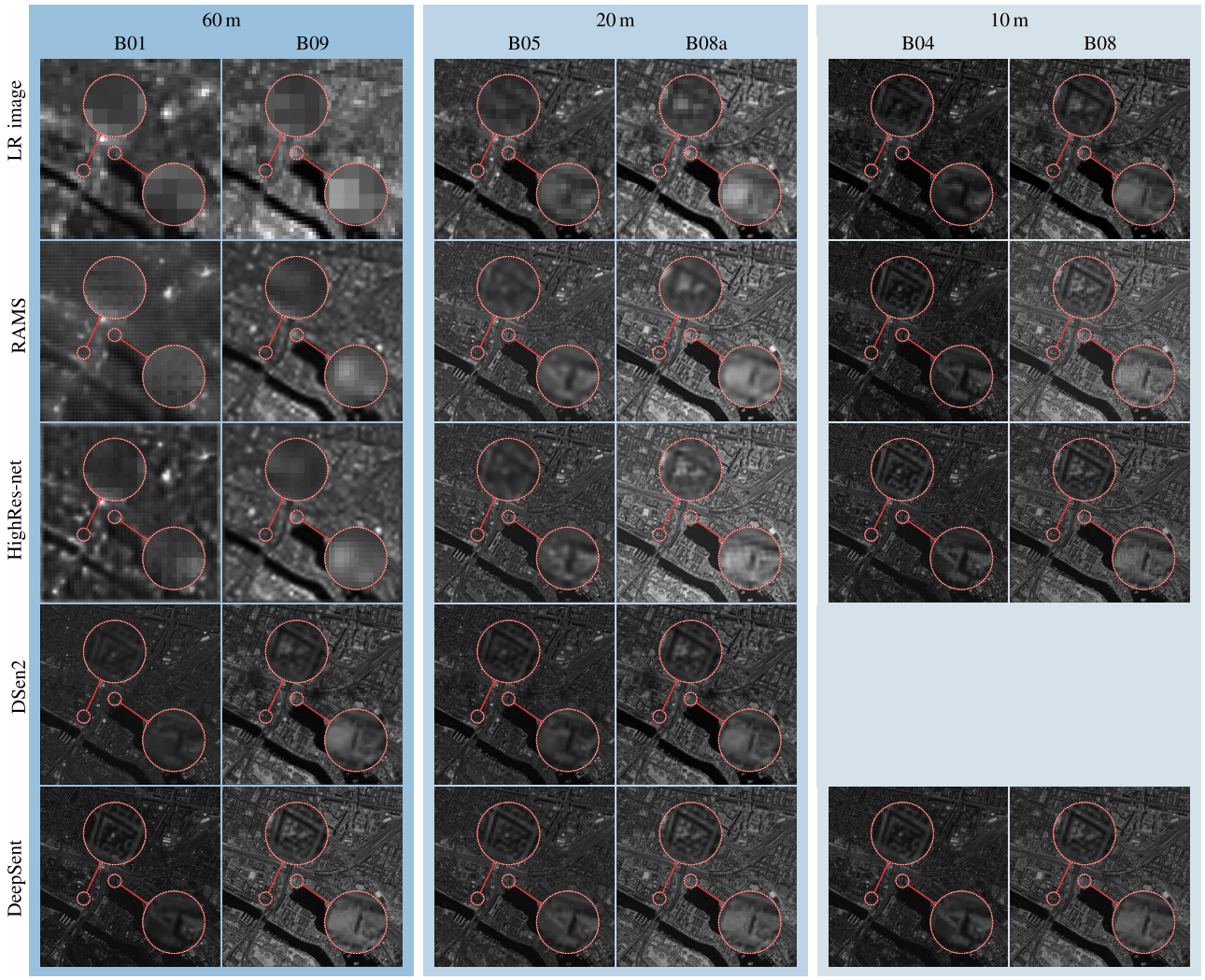

Fig. 12. Super-resolved images obtained from the real-world S-2 image stacks (from the Org-S2 dataset). The input LR and reconstructed images are rendered for six selected bands of the same scene. DSen2 processes only 20- and 60-m bands, so there is no outcome shown for the 10-m bands.

In Fig. 5, we present an example of the super-resolved $B08$ band[5] images (by a factor of $3\times$) for four different Sim-S2 and Sim-HSI scenes. We can appreciate that the results produced by DeepSent contain more high-frequency details, and they are most similar to the HR references. This can also be seen in Fig. 6, where we show a color image composed from the

[5]The remaining bands for these scenes are presented in our Supplementary Material.

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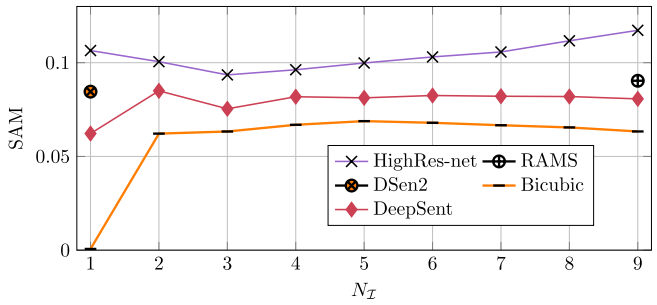

Fig. 13. SAM scores depending on the number of input LR images for reconstruction performed from original S-2 images.

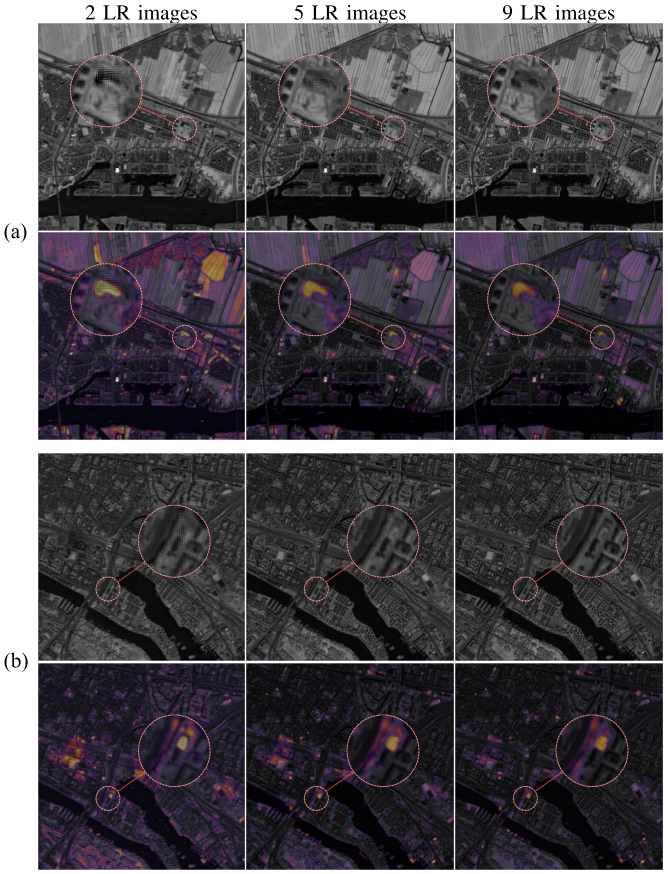

Fig. 14. (a) and (b) Two scenes super-resolved by DeepSent from different numbers of real-world LR images (from Org-S2). The top rows for each scene show the reconstruction outcome, and the bottom rows depict the artifact heat maps superimposed on the outcomes—the brighter (yellowish) pixels are supposed to indicate higher probability of the artifacts. It can be seen that the lower values in the heat maps are observed for using more LR images, and they correspond well with the artifacts intensiveness observed in the top rows.

$B02$, $B03$, and $B04$ bands [for the same scene as in Fig. 5(d)]. It is worth noting that the colors are well reconstructed with DeepSent, which is coherent with the low SAM values reported in Table I. In Fig. 7, we present the outcome of reconstructing the $B06$ (20-m GSD) and $B09$ (60-m GSD) bands [for the same scene as in Fig. 5(a)]. Here, our model renders images 6× and 18× larger than the LR inputs, which is also achieved with the $\text{DSen2}_{\text{HRn}}$ and DSen2_{R} methods; however, their outcome is of substantially lower quality than DeepSent. The remaining techniques operate with lower magnification

TABLE II

RECONSTRUCTION ACCURACY SCORES OBTAINED FOR THE REAL-WORLD IMAGES FROM THE MUS2 BENCHMARK (10-m BANDS). THE BEST RESULTS IN EACH GROUP ARE BOLDFACED, AND THE SECOND BEST ARE UNDERLINED

	cPSNR	cSSIM	LPIPS	$\mathcal{B}$
Bicubic	**30.30**	**0.8168**	0.4220	1
RAMS	29.91	0.7892	<u>0.3186</u>	<u>0.9411</u>
HighRes-Net	29.45	0.7673	0.3212	0.9707
DeepSent	<u>30.15</u>	<u>0.7903</u>	**0.2967**	**0.9179**

TABLE III

NIQE SCORES (THE LOWER, THE BETTER) AVERAGED ACROSS ALL REAL-WORLD PREDICTIONS AND AGGREGATED BY RESOLUTION OF THE INPUT BANDS. THE BEST RESULTS ARE BOLDFACED, AND THE SECOND BEST ARE UNDERLINED

Bands →	60 m	20 m	10 m
LR images	10.71	8.96	7.52
RAMS	10.64	<u>6.67</u>	<u>6.58</u>
HighRes-net	9.23	7.38	6.92
DSen2	<u>7.50</u>	6.87	—
DeepSent	**6.98**	**6.32**	**6.53**

ratios, which results in the fine-grained image details being much worse reconstructed. Finally, Figs. 8 and 9 depict the absolute difference maps with respect to the HR reference image. Black pixels indicate the zero error, whereas the yellow ones annotate the maximal difference. The maps associated with DeepSent are significantly darker, which indicates lower reconstruction errors.

C. Validation Results for Real-World S-2 Data

The capabilities of DeepSent and other techniques for super-resolving real-world images were validated with Org-S2 and MuS2 images. The former is not coupled with any HR reference, and for the latter, HR references are provided for the 10-m S-2 bands. In Table II, we present the quantitative scores obtained for the MuS2 dataset. As argued in [33], cPSNR and cSSIM are not sufficiently robust in this case due to substantial differences between LR and HR images resulting from the fact that they were acquired using different satellites. LPIPS is much more informative here, along with the balanced score ($\mathcal{B}$), specified in the MuS2 benchmark protocol. $\mathcal{B}$ normalizes each score (per each band) with the result obtained using bicubic interpolation (the smaller, the better), as originally proposed in [48]. Distribution of these scores is presented in Fig. 10—DeepSent is slightly better than RAMS, and the difference is significant in the statistical sense (though the differences are not significant for band $B04$ between DeepSent and HighRes-net, and for band $B08$ between DeepSent and RAMS). Qualitative results are presented in Fig. 11. For the $B08$ band [Fig. 11(a)], it can be seen that DeepSent reconstructs the image details that are quite close to the HR reference (it is apparent that there is substantial information gain compared with the input LR image and the outcome of bicubic interpolation). Although we can observe some HR artifacts on the edges, they are much smaller than for RAMS and HighRes-net. An interesting observation can also be made from the color image [Fig. 11(b)] composed of $B02$, $B03$, and $B04$ bands. In this case, the

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TABLE IV

RECONSTRUCTION QUALITY FOR THE 10-m BANDS OBTAINED USING DEEPSENT FROM VARIOUS SETS OF INPUT BANDS ($\boldsymbol{B}_{\text{in}}$) AND A CONSTANT NUMBER OF NINE TEMPORAL IMAGES PER SCENE. THE NUMBER OF BANDS ($|\boldsymbol{B}_{\text{in}}|$) IS PROVIDED IN BRACKETS FOR EACH GROUP. THE BEST RESULTS ARE BOLDFACED, AND THE SECOND BEST ARE UNDERLINED

Metric→		cPSNR				cSSIM			
Input bands ($\boldsymbol{B}_{in}$)	$\lvert\boldsymbol{B}_{in}\rvert$	B02	B03	B04	B08	B02	B03	B04	B08
Only 10 m input bands									
Single band	(1)	42.83	42.00	40.30	35.88	0.9412	0.9308	0.9293	0.8854
$\boldsymbol{B}_{10\text{m}}$	(4)	44.90	43.39	41.02	36.44	0.9574	0.9435	0.9433	0.9049
10 m bands excluded									
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B02}\}$	(11)	—	35.59	33.78	28.17	—	0.8202	0.7800	0.5981
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B03}\}$	(11)	41.15	—	36.12	28.47	0.9285	—	0.8518	0.6110
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B04}\}$	(11)	42.89	41.13	—	28.77	0.9470	0.9328	—	0.6247
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B08}\}$	(11)	45.44	44.04	41.64	—	0.9728	0.9673	0.9542	—
20 m bands excluded									
$\boldsymbol{B}_{\text{all}} \setminus \boldsymbol{B}_{20\text{m}}$	(6)	45.67	44.52	41.64	37.10	0.9745	0.9705	0.9540	0.9141
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B05}\}$	(11)	46.01	44.87	42.04	37.34	0.9767	0.9726	0.9583	0.9200
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B06}\}$	(11)	46.61	45.56	42.64	37.62	0.9791	0.9759	0.9622	0.9235
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B07}\}$	(11)	46.69	45.63	42.69	37.74	0.9793	0.9762	0.9624	0.9247
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B08a}\}$	(11)	46.90	45.82	42.87	37.80	0.9803	0.9772	0.9639	0.9256
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B11}\}$	(11)	45.12	43.93	41.17	36.97	0.9720	0.9668	0.9503	0.9133
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B12}\}$	(11)	45.79	44.66	41.86	37.24	0.9756	0.9715	0.9568	0.9186
60 m bands excluded									
$\boldsymbol{B}_{\text{all}} \setminus \boldsymbol{B}_{60\text{m}}$	(10)	46.06	44.99	42.11	37.20	0.9760	0.9727	0.9577	0.9138
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B01}\}$	(11)	46.89	45.77	42.82	37.75	0.9801	0.9769	<u>0.9636</u>	0.9252
$\boldsymbol{B}_{\text{all}} \setminus \{\text{B09}\}$	(11)	<u>46.97</u>	<u>45.91</u>	<u>42.97</u>	<u>37.82</u>	<u>0.9808</u>	<u>0.9779</u>	**0.9650**	<u>0.9261</u>
All bands									
$\boldsymbol{B}_{\text{all}}$	(12)	**47.11**	**46.02**	**43.06**	**37.98**	**0.9810**	**0.9782**	**0.9650**	**0.9264**

Earth surface in some of LR images is partially occluded with clouds, which affects the result of RAMS and HighRes-net—the clouds are not filtered out, and heavy artifacts can be noticed in these regions. Clearly, DeepSent demonstrates much higher robustness in this case, which may result from combining the multitemporal and multispectral data fusion, and the reconstructed HR information is quite close to the provided reference.

An example of the reconstruction outcome for an image from the Org-S2 dataset is presented in Fig. 12. Considering the 20- and 60-m bands, DeepSent generates images with much more HR details compared with other models, primarily due to its higher upsampling capabilities for these bands. For the 10-m bands, the results generated by DeepSent are comparable with those produced by RAMS and HighRes-net models in terms of the visible details (such as building shapes or streets); however, the latter is heavily contaminated with HR artifacts, which affect the visual quality. This observation is confirmed with the NIQE [90] reported in Table III (the lower, the better). We focus on the problem of the aforementioned artifacts later in this section. Importantly, DeepSent super-resolves the images for all the bands to the resolution of 3.3-m (nominal) GSD, and it can be seen that their detail level is consistent due to the fusion in the spectral dimension and visibly higher than that observed in the input LR images—the gain compared with the original 10-m bands is attributed to the temporal fusion.

Although, for the Org-S2 dataset, we do not have any HR reference, we decided to verify whether the reconstruction preserves the radiometric characteristics. This is particularly important for the spectral fusion techniques (i.e., DeepSent and DSen2) to reassure that they do not copy the spectral properties of 10-m bands while exploiting them to enlarge the bands of lower spatial resolution. To do so, we measure SAM between each set of output images and their corresponding LR counterparts. As the spectral characteristics vary between the input images captured at different times (mainly due to the vegetation), we select the input image, whose spectral distance to the reconstruction outcome is the smallest. Since SAM is calculated in a pixelwise manner, it requires a constant size of all images within a stack. Thus, we downsample both input and output images to the resolution of the input 60-m bands. This downsampling operation also serves as a naive registration by minimizing the subpixel shifts between super-resolved images of different bands. In Fig. 13, we report how the number of input images affects the SAM scores for all investigated algorithms. We present the results obtained for the bicubic interpolation, which can be treated as the SAM lower bound, as it does not modify the spectral information—the increase in SAM when more LR images are used results from the differences among them (the plot shows the difference between the averaged bicubically upsampled image and the most similar input). It can be seen that the results retrieved with DeepSent are slightly worse than this lower bound baseline, and it outperforms other models in terms of maintaining the spectral relations. Also, it preserves almost a constant value when more LR images are included into the input entry.

An important problem for real-world MISR lies in the HR artifacts, which are usually not observed for the simulated data. In Fig. 12, we can observe significant differences in the brightness between the LR images and the corresponding reconstructed ones. In the case of MISR, they result from fusing information provided by multiple input images acquired under inherently different conditions, hence varying in terms of their brightness. Moreover, the time difference between each acquisition implies deviations in local information contained in multiple regions of the very same scene. As we noticed, such content-related variations have a significant negative

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impact on the reconstruction quality. The regions described by inconsistent local information across the time domain have visible artifacts after the SR reconstruction, mainly in the areas containing HR details. To verify our observations, we reconstructed each real-world scene for a variable number of LR images and qualitatively compared the results. In addition, we created the heat maps of estimated probability that some artifacts will occur, by calculating the standard deviation for each pixel in the stack of band-specific temporal images, and averaged them across all bands. From Fig. 14, depicting the SR results and superimposed heat maps, we can see that such artifacts mainly appear in the areas of significant variance and propagate around them. However, as we add more images to the stack, the standard deviation of the local regions diminishes, and the artifacts become less visible. This shows the importance of how the input data are selected for reconstruction, and it explains less intensive artifacts observed for DeepSent than for other techniques, as they exploit less data for the reconstruction.

D. Ablation Study

The goal of our ablation study was to investigate how the information fusion performed in the temporal and spectral domains influences the final outcome. It is composed of two major parts, in which the following hold: 1) we limit the temporal information (by reducing the number of temporal observations presented for reconstruction) to understand the importance of information fusion in the temporal domain and 2) we reduce the spectral information by excluding individual bands (or their subsets) from the input data to verify their usefulness during reconstruction and impact of the information fusion in the spectral domain.

DeepSent architecture (Fig. 1) allows for selecting its forward paths dynamically based on the bands received at its input, as our recursive fusion blocks (at all fusion levels) do not produce any output, if no input is provided. Therefore, the forward pass may be effectively executed with specific parts of the network being switched on or off. In addition, such recursive blocks support feeding the model with different numbers of LR images for each band independently. Given these flexibility features of DeepSent, we could appropriately design the ablation study to understand the abilities of our architecture to exploit different fusion types and the range of the input data.

To verify the capabilities of our network to fuse multitemporal information (and its influence on the SR outcome), we quantified the quality of reconstruction performed from a different number of LR images (from one to nine), for the Sim-S2 dataset. In Fig. 15, we compare the results to other methods tested with the same approach—DSen2 and RAMS are visualized as single points, because they require a fixed number of one and nine LR images, respectively. When SR is performed from a single image (hence without multitemporal data fusion), DSen2 retrieves higher scores (for 20- and 60-m bands) than DeepSent, but for $N_{\mathcal{I}} > 2$, the fusion in the temporal domain performed with our method allows for better reconstruction—this is not the case for RAMS and HighRes-net, which do not benefit from the spectral fusion. We can

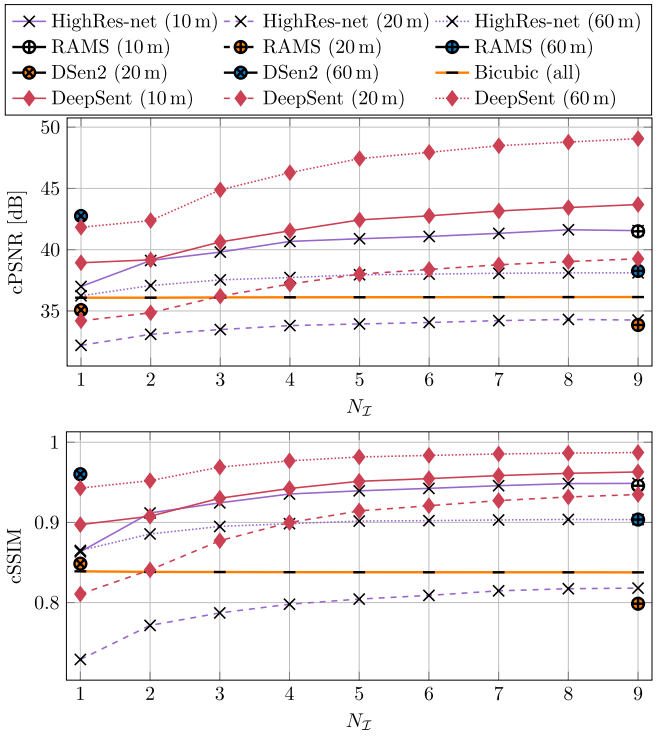

Fig. 15. Reconstruction quality depending on the number of input LR images, obtained with different techniques for 10-, 20-, and 60-m bands (for bicubic interpolation, we show an averaged score for all the bands, as it little depends on $N_{\mathcal{I}}$ for all the bands).

appreciate that the metric values improve, as we increase the number of input images for both DeepSent and HighRes-net. However, our model benefits from it the most, as each additional LR image provides significantly more information, which translates into a greater increase in the quality metrics. In all cases, the improvements in both metrics (and all bands) are statistically significant for DeepSent.

To analyze the spectral fusion capabilities of DeepSent and usefulness of individual spectral bands, we compared the reconstruction quality for the 10-m bands, performed from the following subsets of the input bands: 1) a single 10-m band that is to be reconstructed (thus without any spectral fusion); 2) all four 10-m bands to perform spectral fusion of bands having the same spatial resolution ($\boldsymbol{B}_{10\,\mathrm{m}}$); 3) the 10- and 60-m bands, without the 20-m bands ($\boldsymbol{B}_{\mathrm{all}} \setminus \boldsymbol{B}_{20\,\mathrm{m}}$); 4) the 10- and 20-m bands, without the 60-m bands ($\boldsymbol{B}_{\mathrm{all}} \setminus \boldsymbol{B}_{60\,\mathrm{m}}$); and 5) all of the 12 S-2 bands ($\boldsymbol{B}_{\mathrm{all}}$). In addition to that, we verified the reconstruction quality performed from 11 spectral bands after excluding each of the individual spectral bands (this leads to 12 variants, each of which has a different band excluded). The results are reported in Table IV. It can be seen that the best results are obtained using all the bands, and excluding any single band (or multiple bands) deteriorates the scores. The $B09$ band (60-m GSD) is the least helpful; however, eliminating it from the input bands also slightly decreases the quality scores (except for cSSIM for $B04$—here, the score remained unaffected). As expected, the 60-m bands are less helpful than the 20-m ones, and excluding even one of the 10-m bands severely affects the reconstruction quality (with the $B02$, $B03$,

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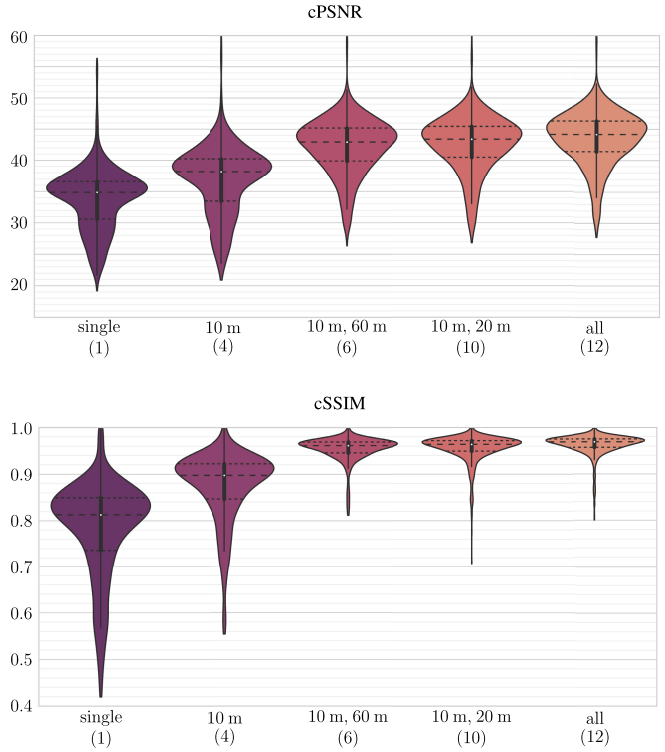

Fig. 16. Distribution of the quality metrics obtained over the test set for 10-m bands with different combinations of input bands: a single band that is to be reconstructed, all 10-m bands ($\boldsymbol{B}_{10\,\mathrm{m}}$), 10- and 60-m bands ($\boldsymbol{B}_{\mathrm{all}} \backslash \boldsymbol{B}_{20\,\mathrm{m}}$), 10- and 20-m bands ($\boldsymbol{B}_{\mathrm{all}} \backslash \boldsymbol{B}_{60\,\mathrm{m}}$), and all the bands ($\boldsymbol{B}_{\mathrm{all}}$). The values reported in parentheses indicate the number of provided bands. The dashed lines indicate the median and quantile values.

and $B04$ bands being more valuable than the NIR $B08$ band). An important conclusion of this analysis is that the information from the lower resolution bands (20 and 60 m) is beneficial to reconstruct the 10-m bands, even though they contain less details than the four 10-m bands. In Fig. 16, we show the distributions of the scores obtained for the aforementioned five subsets of the input bands. They clearly demonstrate that the increase in the amount of spectral information enhances the scores (the improvements are statistically significant for all pairs of investigated configurations and for all bands).

V. Conclusion

In this article, we introduced DeepSent—a new network designed for super-resolving multispectral S-2 images that realizes information fusion in the spectral and temporal dimensions. Our main contribution consists in combining the information provided by multiple images captured at different times along with multiple spectral bands. To the best of our knowledge, such collation of temporal and spectral information fusion has not been reported in the field of SR so far. A unique feature of DeepSent is that a single model is capable of super-resolving all of the S-2 bands to the uniform resolution of 3.3-m nominal GSD, while preserving the spectral relations between the bands. Quantitative and qualitative results for both the simulated data, as well as for the original Level-2A S-2 images, clearly indicate that DeepSent outperforms other state-of-the-art solutions, which incorporate only one type of fusion (i.e., RAMS and HighRes-net for temporal, and DSen2 for spectral fusion). The reconstruction quality is also higher than when the single-fusion techniques are combined in a sequential manner (i.e., DSen2 followed by RAMS or HighRes-net). In the case of super-resolving real-world data, we have demonstrated that the outcome retrieved by DeepSent is characterized with much less artifacts than the outcomes obtained with other state-of-the-art methods. We have also provided evidence that the artifacts result from inconsistent local information contained in an image time series, which may be a helpful hint for other researchers working in this field.

The results reported in this article constitute a solid basis for further research. First, even though DeepSent allows for reconstructing all the spectral bands using the same earlier-trained model, they have to be retrieved one after another by passing appropriate input band through the residual connection—reconstructing the whole MSI in one go would be much more convenient; however, it is challenging due to memory issues. Furthermore, the residual connection transmits an averaged image of the whole temporal series, making it impossible to select a particular image in the series that should be reconstructed. Even though the super-resolved images preserve the spectral characteristics, the ability to pick a single image in the series that should be reconstructed using the remaining images in that series would be an extremely useful feature of high practical importance. Finally, although DeepSent offers a lower level of artifacts than other methods when fed with original S-2 images, this remains an open research problem worthwhile being addressed in the future. One of the possibilities is to focus on developing more advanced methods of generating simulated datasets for training that would better reflect the real-world data characteristics. Another possibility would be to enhance the recursive fusion blocks with attention mechanism to select most valuable features that do not lead to visual artifacts.

Last but not least, the proposed framework could be exploited for solving specific image analysis tasks, which require higher spatial resolution than that of the original S-2 products. Such application-related evaluation, being in line with the current trends in real-world SR, would also help in determining future research pathways that could lead to novel practical applications of SR products. Even though most of the emerging SR solutions are evaluated relying on HR reference data, elaborating new task-oriented SR benchmarks is an important step toward solidifying the research on SR and enhancing the robustness of the existing techniques. With our contribution, we support that goal by making our source code, training data, and trained models publicly available—this will ensure the reproducibility and allow our method to be validated in numerous practical future applications.

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